Planar Camera Calibration with a Mobile Phone

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This experiment estimates a phone camera's intrinsic matrix and lens distortion from 19 views of a planar checkerboard. All 19 images yielded the required 40 internal corners. OpenCV calibration returned an overall in-sample reprojection RMS of 0.942 pixels at 4032 by 3024 resolution. The report documents the actual inputs, fitted parameters, image evidence, and a one-view sensitivity check. The board was displayed on a laptop screen; its stated 25 mm square size was supplied for this experiment and was not independently measured here.

Aim and data

The aim was to recover the intrinsic matrix K and distortion coefficients of one phone camera using Zhang's planar calibration method [Zhang, 1999].[1] Nineteen HEIC photographs were converted to JPEG with `heif-convert` at quality 95. Every image is 4032×3024 pixels. The pattern contains 9 by 6 squares, hence $8 \times 5 = 40$ internal corners. The square length entered in the program was $s = 0.025$ m. The checkerboard was shown on a laptop display rather than printed. The Labo handout mentions a 3 cm board; that dimension was not used for this screen pattern.

[1] The phone model recorded in the converted image EXIF is Apple iPhone 14 Pro. The individual lens is not identified by that field.

Method

For each accepted photograph, planar object points were defined as $M_{ij} = (is, js, 0)^{\mathsf{T}}$. Image corners were detected with OpenCV and refined to subpixel precision. The SB detector was available as a fallback. With a different board pose in each view, the pinhole model is

$$\lambda \widetilde{\boldsymbol{m}} = K[\, R \mid \boldsymbol{t} \,]\widetilde{\boldsymbol{M}}, \qquad K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}. \tag{1}$$

OpenCV's `calibrateCamera` [OpenCV] jointly fitted K, the five reported distortion coefficients, and a pose for each image. For N accepted corners, the reported residual was independently recomputed as

$$E_{\mathrm{RMS}} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \| \boldsymbol{m}_i - \widehat{\boldsymbol{m}}_i \|_2^2}. \tag{2}$$

Here $\widehat{\boldsymbol{m}}_i$ is the point returned by `projectPoints`. This is an in-sample fit statistic, not a measurement against an external reference.

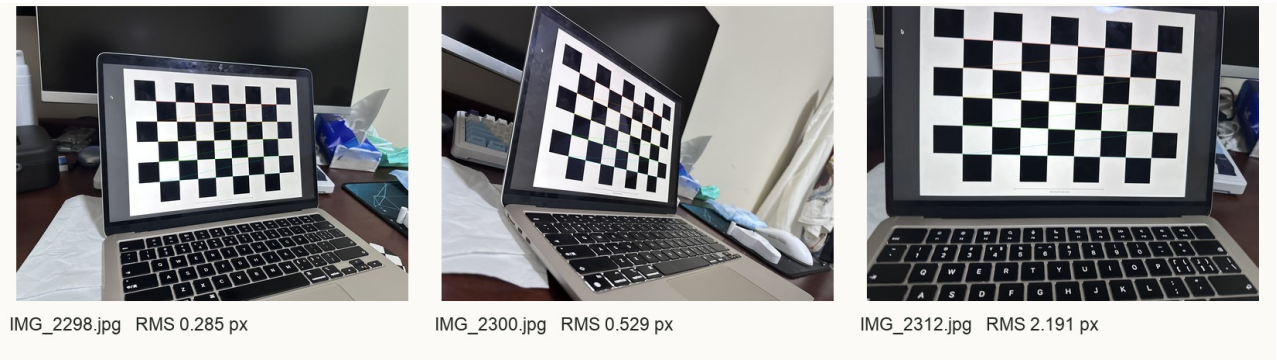


Figure 1: Three accepted images with detected internal corners. The first two illustrate different viewing conditions; `IMG_2312` has the largest per-image reprojection error. Figure labels give each view's RMS in pixels.

Results

All 19/19 photographs were accepted. The fitted intrinsic matrix is

$$K = \begin{bmatrix} 2859.815 & 0 & 2014.596 \\ 0 & 2856.387 & 1523.779 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{pixels.} \tag{3}$$

The coefficient vector in OpenCV order $(k_1, k_2, p_1, p_2, k_3)$ is

$$(0.201096,\ -0.762019,\ -0.000094,\ 0.000053,\ 0.657344). \tag{4}$$

The overall RMS is 0.942 px. The best individual view, `IMG_2298`, has 0.285 px RMS; `IMG_2312` has 2.191 px. The principal point is close to the image center $(2016, 1512)$, which is a useful plausibility check but not an independent accuracy test.

In a sensitivity check, `IMG_2312` was omitted. Refitting the other 18 photographs gave 0.806 px RMS, with $f_x \approx 2854.3$ px and $f_y \approx 2852.1$ px. The focal-length changes are below 0.2% relative to the 19-view fit. The delivered parameters remain the 19-view result; this comparison does not justify deleting an observation without checking its image.

Interpretation and limits

The images include changes in distance and orientation, which provide constraints beyond a single frontal view. The relatively large higher-order radial coefficients deserve caution: a screen pattern can introduce pixel structure, glare, and interpolation effects, and only one dataset was evaluated. No held-out calibration target or independent physical measurement was used. Because the board's physical square length only sets object-space scale, an error in the stated 25 mm size scales the reported pose translations; it does not by itself change the pixel-valued intrinsic matrix.

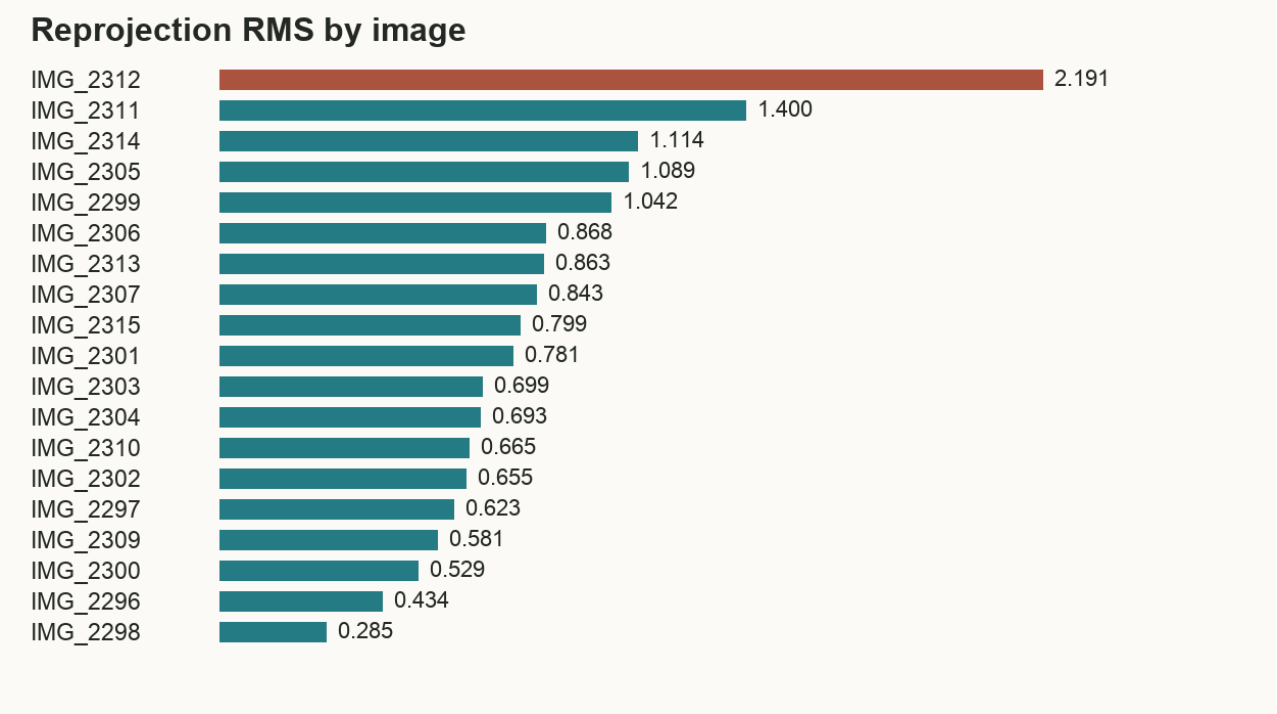


Figure 2: Per-view reprojection RMS for all 19 photographs, sorted from largest to smallest. The highlighted view motivates the sensitivity check below.

Reproduction

The run used Python 3.9.23, NumPy 2.0.2, and OpenCV 4.12.0. From the repository root:

```
python3.9 lab0/code/calibrate_phone.py
-images lab0/data/phone_jpg -square-size-m 0.025
```

The complete per-image results, parameter array, and annotated images are saved under `lab0/results/calibration/`. The report figures were selected from those saved outputs.

References

OpenCV. Camera calibration. `https://docs.opencv.org/4.x/dc/dbb/tutorial_py_calibration.html`. Official OpenCV tutorial.

Zhengyou Zhang. Flexible camera calibration by viewing a plane from unknown orientations. In *Proceedings of the IEEE International Conference on Computer Vision*, 1999.